When the Horizon Spins: Kerr Rotation in Clock Language

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Abstract

The Kerr solution describes an uncharged, rotating black hole by its mass M and angular momentum J. Define the gravitational timescale $\tau_g \equiv GM/c^3$ and a spin timescale $\tau_J \equiv J/(Mc^2)$. Their ratio $\chi \equiv \tau_J/\tau_g = cJ/(GM^2)$ is the signed dimensionless spin. Kerr horizons exist for $|\chi| \leq 1$. In this notation, the Boyer–Lindquist horizon radii divided by c are $\tau_\pm = \tau_g(1 \pm \sigma)$, where $\sigma \equiv \sqrt{1-\chi^2}$. The same parameters express the horizon angular velocity and the semiclassical Hawking temperature, and turn the Kerr Smarr relation into a compact split with fractions σ and $1-\sigma$. We also relate the horizon area to entropy and its value in bits, and define the irreducible-mass scale. These are exact rearrangements of Kerr geometry and its classical thermodynamic identities, combined with the standard semiclassical temperature formula. The quantities τ_J and $\tau_\pm$ are not proper times read by clocks at the horizon, Ω_H is not the rigid rotation of a material surface, and coordinate convergence of orbit radii at extremality does not by itself mean physical co-location.

Keywords: Kerr black hole, frame dragging, horizon angular velocity, Hawking temperature, Smarr relation, irreducible mass

1 Introduction

The Kerr geometry adds angular momentum to the mass-only Schwarzschild description. Its horizon structure, angular velocity, temperature, area, and mass formula can be expressed using two time-valued parameters: a gravitational timescale and a spin parameter written as a time. This is a change of variables within the established Kerr solution, not a new gravitational law [1, 2].

The purpose of this paper is to derive that notation carefully and keep its interpretation bounded. We distinguish coordinate radii divided by c from proper time, the horizon's

angular velocity from rigid-body rotation, and the Smarr terms from a literal partition into independent energy reservoirs. The Kerr relations are exact within the stated idealization. The Hawking temperature is the standard semiclassical result for a stationary Kerr black hole, normalized with respect to time at infinity [3, 4].

Unless stated otherwise, we consider a four-dimensional, asymptotically flat, uncharged Kerr black hole. We use χ for signed spin and report nonnegative χ in examples aligned with the chosen rotation axis. No weak-field expansion is used.

2 Spin expressed as a time

Let M be the black-hole mass and J its signed angular momentum. Define

$$\tau_g \equiv \frac{GM}{c^3}, \qquad a \equiv \frac{J}{Mc}, \qquad \tau_J \equiv \frac{a}{c} = \frac{J}{Mc^2}. \tag{1}$$

Here a is the usual Kerr spin parameter with units of length; τ_J is that length divided by c. We then define the dimensionless spin

$$\chi \equiv \frac{\tau_J}{\tau_g} = \frac{cJ}{GM^2}. \tag{2}$$

The term "spin time" is used here as a convenient name for τ_J, not as standard terminology. It is a parameter constructed from M and J, not an elapsed time or a clock reading.

The gravitational radius is

$$r_g \equiv \frac{GM}{c^2} = c\tau_g. \tag{3}$$

The Kerr horizon condition is

$$|a| \le r_g, \qquad |\tau_J| \le \tau_g, \qquad |\chi| \le 1. \tag{4}$$

At fixed M, the limiting angular momentum is therefore

$$J_{\max} = \frac{GM^2}{c} = Mc^2\tau_g. \tag{5}$$

If $|\chi| > 1$, the vacuum Kerr metric has no event horizon. An object whose assumed M and J give this ratio cannot be described as a Kerr black hole without changing those inputs.

Using the unreduced Planck mass $M_P \equiv \sqrt{\hbar c/G}$, the same bound can be written in units of action:

$$\frac{J_{\max}}{\hbar} = \frac{GM^2}{\hbar c} = \left(\frac{M}{M_P}\right)^2. \tag{6}$$

This is the classical Kerr limit expressed as a dimensionless ratio; it does not assert that black-hole angular momentum is quantized in steps of $\hbar$. The mass–area identity $M\ell_P^2 = \hbar\tau_g$ remains valid for Kerr, because it depends on M and the definitions of ℓ_P and τ_g, not on J.

3 The two horizon scales and the ergosurface

The two Kerr horizon radii in Boyer–Lindquist coordinates are [2]

$$r_\pm = r_g \pm \sqrt{r_g^2 - a^2}. \tag{7}$$

Introduce the nonnegative quantity

$$\sigma \equiv \sqrt{1 - \chi^2}. \tag{8}$$

Dividing Eq. (7) by c yields the coordinate-radius scales

$$\tau_\pm \equiv \frac{r_\pm}{c} = \tau_g \pm \sqrt{\tau_g^2 - \tau_J^2} = \tau_g(1 \pm \sigma). \tag{9}$$

The notation $\tau_\pm$ does not denote proper time measured by an observer at either horizon. It is the Boyer–Lindquist radial coordinate expressed in time units. For $0 < |\chi| < 1$, the inner root is the Cauchy horizon of the ideal subextremal Kerr solution; this does not assert that the inner structure is stable under perturbations. At $\chi = 0$, the formal root $r_- = 0$ is the Schwarzschild singularity, not an inner horizon. At extremality, the two roots coalesce into a degenerate horizon.

The two roots satisfy

$$(\tau_+ - \tau_g)^2 + \tau_J^2 = \tau_g^2. \tag{10}$$

For zero spin, $\tau_+ = 2\tau_g$ and $\tau_- = 0$. In the extremal limit $|\chi| \to 1$, the roots coincide at $\tau_+ = \tau_- = \tau_g$. This is a degeneracy of the horizon roots in the Kerr solution, not two physical clocks meeting.

Rotation also distinguishes the outer horizon from the outer static limit. The latter, which bounds the ergoregion in Boyer–Lindquist coordinates, is

$$r_{\text{ergo}}(\theta) = r_g + \sqrt{r_g^2 - a^2\cos^2\theta}. \tag{11}$$

The static limit generally lies outside the event horizon except at the rotation axis. The relevant stationary metric sector can be written as [2]

$$\mathrm{d}s^2 = -\alpha^2 c^2 \mathrm{d}t^2 + g_{\phi\phi}(\mathrm{d}\phi - \omega\,\mathrm{d}t)^2 + g_{rr}\mathrm{d}r^2 + g_{\theta\theta}\mathrm{d}\theta^2, \tag{12}$$

where α is the Kerr lapse, ω is the angular shift or frame-dragging rate, and $g_{\phi\phi}$ is the azimuthal metric coefficient. A timelike circular observer with $\mathrm{d}\phi/\mathrm{d}t = \Omega$ therefore has

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{\alpha^2 - \frac{g_{\phi\phi}}{c^2}(\Omega - \omega)^2}. \tag{13}$$

A locally nonrotating observer has $\Omega = \omega$ and hence $\mathrm{d}\tau/\mathrm{d}t = \alpha$. This observer is generally not fixed in Boyer–Lindquist azimuth. Within the ergoregion, the fixed-coordinate choice $\Omega = 0$ is not timelike, so no observer can remain at fixed spatial coordinates there. Frame dragging is not the rigid rotation of a material horizon.

4 Horizon angular velocity

The angular velocity of the horizon generator, defined relative to the time and azimuthal coordinates normalized at infinity, is [2, 3]

$$\Omega_H = \frac{ac}{r_+^2 + a^2} = \frac{\tau_J}{\tau_+^2 + \tau_J^2} = \frac{\chi}{2\tau_g(1+\sigma)}. \tag{14}$$

Its sign follows the sign of J. For nonzero spin, a coordinate rotation period may be defined as $P_H \equiv 2\pi/|\Omega_H|$. This period is distinct from the spin parameter τ_J; neither quantity is the proper period of a material surface rotating at the horizon.

At zero spin, $\Omega_H = 0$. At extremality,

$$|\Omega_H| \longrightarrow \frac{1}{2\tau_g}. \tag{15}$$

For a solar-mass scale, $\tau_g \simeq 4.9255\ \mu$s [11], so the extremal angular velocity is approximately $1.0151 \times 10^5\ \mathrm{rad\,s^{-1}}$, or 16.156 kHz when expressed as cycles per second. The radian-per-second value is an angular velocity; the hertz value is its division by 2π.

5 Surface gravity and the Hawking temperature

For a subextremal Kerr black hole, the surface gravity is [3]

$$\kappa_H = \frac{c^2(r_+ - r_-)}{2(r_+^2 + a^2)} = \frac{c\sigma}{2\tau_g(1+\sigma)}. \tag{16}$$

The standard semiclassical Hawking temperature associated with this surface gravity is [4]

$$k_{\mathrm{B}} T_H = \frac{\hbar \kappa_H}{2\pi c} = \frac{\hbar \sigma}{4\pi \tau_g (1+\sigma)}. \tag{17}$$

This temperature is normalized relative to time at infinity. It is not a local temperature measured by an observer hovering at the horizon. Quantum-field and back-reaction corrections beyond the standard semiclassical expression are outside this paper.

Define the Schwarzschild temperature at the same mass by

$$T_0 \equiv \frac{\hbar}{8\pi k_{\mathrm{B}} \tau_g}. \tag{18}$$

The spin dependence is then

$$\frac{T_H}{T_0} = \frac{2\sigma}{1+\sigma}. \tag{19}$$

At zero spin this ratio is one; as $|\chi| \to 1$, the temperature tends to zero in the classical extremal limit. The formula describes the Kerr family and its limit; it does not establish that a real astrophysical black hole reaches exact extremality.

6 The Kerr Smarr relation

For a quasistatic change within the Kerr family, the first law is

$$\mathrm{d}(Mc^2) = T_H \, \mathrm{d}S_{\mathrm{BH}} + \Omega_H \, \mathrm{d}J. \tag{20}$$

The Smarr mass formula is a separate relation among the state variables [5]:

$$Mc^2 = 2T_H S_{\mathrm{BH}} + 2\Omega_H J. \tag{21}$$

Using Eqs. (2)–(17) and the Kerr area entropy gives the exact ratios

$$\frac{2T_H S_{\mathrm{BH}}}{Mc^2} = \sigma, \qquad \frac{2\Omega_H J}{Mc^2} = 1 - \sigma. \tag{22}$$

At zero spin the rotational term vanishes and Eq. (21) reduces to the Schwarzschild relation $Mc^2 = 2T_H S_{\rm BH}$. At extremality the Smarr thermal term tends to zero and the rotational term equals Mc^2.

The two terms are the contributions in the Smarr identity. They should not be read as independently stored energy reservoirs, and the rotational Smarr term $2\Omega_H J$ is not the same quantity as the maximum rotational energy that can be extracted in a reversible process.

7 Area, entropy, and bits

The Kerr horizon area is

$$A = 4\pi(r_+^2 + a^2) = 8\pi c^2 \tau_g \tau_+. \tag{23}$$

Define its area in Planck-area units by

$$n_A \equiv \frac{A}{\ell_P^2}, \qquad \ell_P^2 \equiv \frac{\hbar G}{c^3}. \tag{24}$$

The Bekenstein–Hawking area law and its expression in bits are [6]

$$\frac{S_{\rm BH}}{k_{\rm B}} = \frac{n_A}{4}, \qquad n_{\rm bit} \equiv \frac{S_{\rm BH}}{k_{\rm B} \ln 2} = \frac{n_A}{4 \ln 2}. \tag{25}$$

Here n_A is an area ratio, not a count of literal horizon tiles; $n_{\rm bit}$ expresses entropy in bit units and does not, by itself, identify physical memory elements.

Substituting Eq. (25) into the thermal part of Eq. (21) gives

$$Mc^2 - 2\Omega_H J = 2T_H S_{\rm BH} = \frac{1}{2} n_A k_{\rm B} T_H. \tag{26}$$

This is an algebraic combination of the area law and Kerr Smarr relation. It does not imply a microscopic equipartition mechanism on the horizon. The mass-only scales from the preceding manuscript also carry over to Kerr: with $I_* \equiv M\ell_P^2$, $\omega_g \equiv 1/\tau_g$, $\tilde{\tau}_C \equiv \hbar/(Mc^2)$, and $t_P \equiv \sqrt{\hbar G/c^5}$, one has

$$I_* \omega_g = \frac{M\ell_P^2}{\tau_g} = \hbar, \qquad \tau_g \tilde{\tau}_C = t_P^2, \qquad E_* \equiv \frac{1}{2} I_* \omega_g^2 = \frac{\hbar}{2\tau_g}. \tag{27}$$

Here I_* remains a defined inertia scale and ω_g an inverse-time scale, not the black hole's angular velocity. For the Kerr Hawking temperature, the formal scale obeys

$$\frac{E_*}{k_{\rm B} T_H} = 2\pi \frac{1+\sigma}{\sigma}, \qquad \sigma > 0. \tag{28}$$

This ratio is 4π only at zero spin ($\sigma = 1$); it increases with the magnitude of spin and diverges as $\sigma \to 0^+$, when the extremal temperature tends to zero. Thus the $E_* = 4\pi k_{\rm B} T_H$ identity is specific to Schwarzschild.

8 Irreducible mass and the extractable-spin bound

The irreducible mass is defined by the horizon area through $A = 16\pi G^2 M_{\rm irr}^2/c^4$. Equations (23) and (8) then give [7]

$$M_{\rm irr}^2 = \frac{M^2}{2}(1+\sigma). \tag{29}$$

For zero spin, $M_{\rm irr} = M$. In the extremal limit, $M_{\rm irr} = M/\sqrt{2}$. Under ideal reversible extraction, the mass-energy above the irreducible mass sets the maximum available rotational-energy budget:

$$\frac{E_{\rm rot,max}}{Mc^2} = 1 - \frac{M_{\rm irr}}{M} = 1 - \sqrt{\frac{1+\sigma}{2}}. \tag{30}$$

At extremality this ideal upper bound is $1 - 1/\sqrt{2} \simeq 29.3\%$. It is not a prediction that this fraction will be extracted in an actual astrophysical process. The area theorem and the irreducible-mass concept constrain classical processes under their usual assumptions [3, 7].

9 Dimensionless checks and examples

Table 1 gives representative values of the dimensionless Kerr functions. The entries are ratios, independent of the black-hole mass; the $\chi = 0.998$ row is included for comparison with a particular accretion model, not as the exact Kerr bound.

Table 1: Kerr horizon and temperature scales for selected nonnegative spins. T_0 is the Schwarzschild temperature at the same mass.

χ	σ	τ_+/τ_g	τ_-/τ_g	$\Omega_H \tau_g$	T_H/T_0
0	1	2	0	0	1
0.22	0.97550	1.97550	0.02450	0.05568	0.98760
0.8	0.60000	1.60000	0.40000	0.25000	0.75000
0.998	0.06321	1.06321	0.93679	0.46933	0.11891
1	0	1	1	0.5	0

For a solar-mass scale, $\tau_g \simeq 4.9255$ μs and $T_0 \simeq 61.70$ nK [11]. The helioseismic estimate $J_\odot \simeq 1.92 \times 10^{41}$ kg m^2 s^{-1} corresponds to $\chi \simeq 0.22$ if the Sun's mass and angular momentum are hypothetically assigned to a Kerr black hole [10, 11, 13]. This is an illustrative parameter

comparison, not a model of solar collapse. The actual collapse of a star need not preserve either quantity.

As a contrasting thought experiment, an axial moment-of-inertia model $C_\oplus \simeq 8.0365 \times 10^{37}\ \mathrm{kg\,m^2}$ and the present rotation rate $\Omega_\oplus \simeq 7.292115 \times 10^{-5}\ \mathrm{s^{-1}}$ give $J_\oplus = C_\oplus \Omega_\oplus \simeq 5.86 \times 10^{33}\ \mathrm{kg\,m^2\,s^{-1}}$ [12]. Assigning this angular momentum and the Earth's present mass to a Kerr black hole gives $|\chi| \sim 7.4 \times 10^2$ [11, 13], well above the horizon bound. Such an object would have to shed angular momentum, change mass, or both before it could be represented by a Kerr black hole; this is an illustrative conservation check, not a model of Earth's collapse.

The exact Kerr extremality bound is $|\chi| = 1$. Thorne's often-quoted value near 0.998 arises in a specified radiatively efficient thin-disk accretion model that includes captured radiation [9]. It is not a universal replacement for the Kerr bound and should be presented with those model assumptions.

10 Orbit radii and the extremal limit

For prograde equatorial motion, the Boyer–Lindquist radii of the circular photon orbit and the innermost stable circular orbit approach the horizon radius as extremality is approached [2]. At zero spin, the corresponding radii are $3r_g$ and $6r_g$, while the horizon is at $2r_g$. At extremality, equality of the limiting Boyer–Lindquist radial coordinates does not by itself imply that the orbit and horizon are the same physical location. The spatial relation depends on the time slicing: on a constant Boyer–Lindquist-time slice, the photon orbit and ISCO remain distinct from the horizon; on a horizon-crossing slice, their limiting locations can coincide with the horizon generators [8].

This coordinate subtlety is why the manuscript treats the three radii as a useful comparison of coordinate scales, not as a literal collision of physical surfaces or orbits.

11 Interpretation and limits

The clock-language construction is a reparameterization of Kerr quantities. The spin time τ_J packages J and M into a quantity with units of time; it does not measure a local clock's proper time. The horizon scales $\tau_\pm$ are coordinate radii divided by c, and Ω_H describes the horizon generator relative to the asymptotic frame, not a rigid material rotation. The entropy-in-bits expression is a change of entropy units, not evidence for physical horizon bits.

The geometric statements assume the exact stationary, uncharged Kerr solution. The temperature formula adds the standard semiclassical Hawking relation. Astrophysical accre-

tion limits, collapse examples, and orbit-radius interpretations require their own assumptions and are not universal consequences of the algebra. No new causal mechanism, independent energy reservoir, or quantization of black-hole angular momentum is proposed.

12 Conclusion

Writing $\tau_g = GM/c^3$ and $\tau_J = J/(Mc^2)$ makes the dimensionless Kerr spin the ratio $\chi = \tau_J/\tau_g$. The horizon condition becomes $|\tau_J| \leq \tau_g$, and the coordinate radii divided by c take the form $\tau_\pm = \tau_g(1 \pm \sqrt{1-\chi^2})$. The same variables express the horizon angular velocity, the semiclassical temperature, and the Kerr Smarr relation.

These compact forms are exact rearrangements within their stated domain. Keeping coordinate time scales distinct from proper time, horizon angular velocity distinct from material rotation, and Smarr bookkeeping distinct from extractable rotational energy makes the clock-language bridge useful without overstating its physical meaning.

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